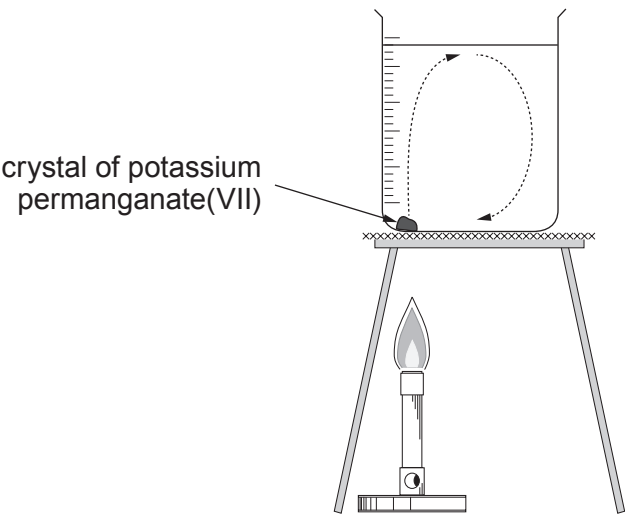


5. A class carries out experiments to study heat transfer in solids and liquids.

- (a) In one experiment, the process of convection is shown by colouring some water with potassium permanganate(VII) crystals in the bottom corner of a beaker. The water is then heated and a clockwise convection current is set up.



Explain why the coloured water moves in the way described above.

[2]

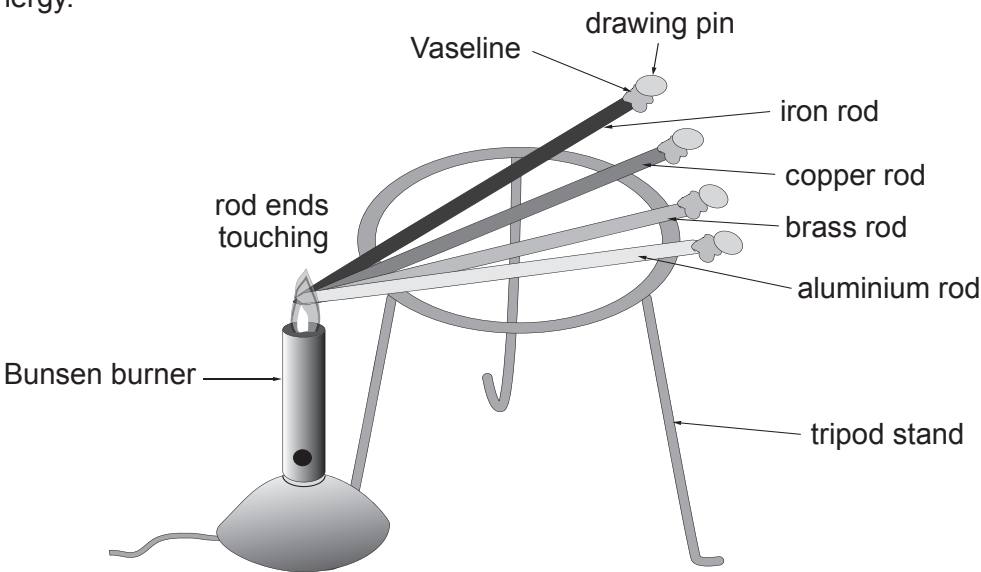
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- (b) In another experiment, four metal rods are tested to compare their ability to conduct heat energy.



Examiner
only

The outcome of the experiment is as follows:

- copper (best conductor)
- aluminium
- brass
- iron (poorest conductor).

Explain in detail, the process of conduction in **copper** in terms of the motion of **particles**.
[6 QER]

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